Pavel Zakharov

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Education

Bielefeld University, Germany, 2026 – present
Faculty of Mathematics, PhD Program

TU Dortmund, Germany, 2023 – 2026
Department of Mathematics, PhD Program

Higher School of Economics, Moscow, Russia, 2021 – 2023
Data Science, Math in Machine Learning, Master Program

Higher School of Economics, Moscow, Russia, 2017 – 2021
Faculty of Computer Science, Applied Mathematics and Informatics, Bachelor Program

Physical-Mathematical School 2007, Moscow, Russia, 2017
Graduated with Golden Medal.

Positions

Bielefeld University, Faculty of Mathematics, 2026 – present
Research Assistant

TU Dortmund, Department of Computer Science, 2023 – 2026
Research Assistant

University of New South Wales, 2025
Research Internship

HSE University, Laboratory of Theoretical Computer Science, 2019 – 2023
Research Assistant

Yandex LLC, 2019 – 2020
Yandex.Routing, Yandex.Adtech, Software engineer

Research

Preprints

Threshold existence for random NAE-k-SAT and random hypergraph 2-colourability, 2026
P. Zakharov.

The chromatic number of a random graph with $o(n^{3/2})$ edges, 2026
A. Coja-Oghlan, M. Isaev, M. Kang, P. Zakharov.

Fluctuations of the Ising free energy on Erdős-Rényi graphs, 2026
A. Coja-Oghlan, D. Kaaser, M. Rolvien, P. Zakharov, K. Zampetakis.

The number of random 2-SAT solutions is asymptotically log-normal 2024
A. Chatterjee, A. Coja-Oghlan, N. Müller, C. Riddlesden, M. Rolvien, P. Zakharov, H. Zhu,
To appear in *Theory of Computing*.

Journal publications

A note on the maximum clique LP relaxation,
P. Zakharov,
Operations Research Letters **66** *(2026) 107425*.

The random k-SAT Gibbs uniqueness threshold revisited,
A. Chatterjee, A. Coja-Oghlan, C. Greenhill, V. Pfenninger, M. Rolvien, P. Zakharov, K. Zampetakis,
Combinatorics, Probability and Computing **35** *(2026) 1–53*.

WalkSAT is linear on random 2-SAT,
P. Berenbrink, A. Coja-Oghlan, C. Cooper, T. Götte, L. Hintze, P. Zakharov,
SIAM Journal on Discrete Mathematics **39 (4)** *(2025) 1939–1952*.

Fractional colourings of random hypergraphs,
D. Shabanov, P. Zakharov,
Russian Mathematical Surveys **78 (6)** *(2023) 183–184*.

Asymptotic normality of the giant component size in a random bipartite graph,
D. Shabanov, P. Zakharov,
Trudy MIPT **15** *(2023) 23–32 (in Russian)*.

On the maximal cut in a random hypergraph,
D. Shabanov, P. Zakharov,
Doklady Mathematics **104** *(2021) 336–339*.

Talks

The 43rd Kolloquium über Kombinatorik, 2026
Berlin, Germany.

Algorithms, Dynamics, and Information Flow in Networks Seminar, 2026
Online.

31st British Combinatorial Conference, 2026
Cardiff, United Kingdom.

Randomness in Computing, 2026
Sheffield, United Kingdom.

Athens-Dortmund-Hamburg Workshop on Graphs, Networks, Optimization, and Statistical Inference, 2026
Corfu, Greece.

Midwinter meeting in Discrete Probability, 2026
Umeå, Sweden.

Algorithmic Decision Theory Reading Group, 2025
Sydney, Australia.

22nd International Conference on Random Structures and Algorithms, 2025
Vienna, Austria.

28th International Conference on Randomization and Computation, 2024
London, United Kingdom.

Algorithms, Dynamics, and Information Flow in Networks Seminar, 2024
Online.

European Conference on Combinatorics, Graph Theory and Applications, 2021
Online.

Selected Teaching

Bielefeld University, Faculty of Mathematics 2026-2027
Graph Theory — Tutoring

TU Dortmund, Department of Computer Science 2026
Complexity Theory — Design of study materials, tutoring

Higher School of Economics, Faculty of Computer Science 2021 – 2024
Theoretical Computer Science — Lecturer, course design
Mathematical Statistics — Design of study materials, tutoring
Algorithms and Data Structures — Design of study materials, tutoring
Theory of Computation — Tutoring
Convex Programming and Approximation Algorithms — Tutoring
Probability Theory — Tutoring

Research grants

Co-investigator on “Extremal properties of random graphs". Russian Foundation for Basic Research and Iran National Science Foundation, no. 20-51-56017 — 2020 – 2022

Co-investigator on “Structural properties of random graphs and hypergraphs". Russian Science Foundation, no. 22-21-00411 — 2021 – 2023

Service and outreach

ADYN Summerschool on Algorithms, Dynamics, and Information Flow in Networks — 2026
Organizing

“Combi meets Stat Physics" reading group, UNSW — 2025
Organizing

Peer reviewing and subreviewing
Annals of Applied Probability, Australasian Journal of Combinatorics, Combinatorica, Random Structures and Algorithms, STACS

Project programme in mathematics & theoretical computer science, “Sirius" Educational Center — 2022
Mentor

Mathematics summer school “Kostroma open" — 2017
Invited teacher

Other

Ilya Segalovich scholarship — 2021-2022
Awarded by Yandex, €10K